

HCI Research in Sensitive Settings: Lessons Learned from Technology Design and Ethical Challenges in Dementia

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ABSTRACT

In this course, we will share concrete guidelines for HCI research in sensitive settings based on our research experience in dementia. We share how these lessons learned apply in sensitive settings regarding how designing for and with people with dementia has general implications for design and computing. First, we will provide real-world insights on designing technologies in sensitive settings by analyzing example cases from academia and industry. Next, we instruct designers and researchers in practical ethical conduct with stakeholders through hands-on exercises, such as facilitating informed consent and modifying common co-design activities for sensitive contexts.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Field studies.

KEYWORDS

Course, Dementia, Design, Ethics, Sensitive Settings

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1 BENEFITS

Over the years, there have been numerous research efforts in HCI to include vulnerable users in sensitive settings in the design process [12]. For example, there has been an emerging body of research in HCI on how technology and design can support the quality of life

of people with dementia [1]. These research efforts have explored the challenges and opportunities for working closely with people with dementia in the design process [10]. However, design research in sensitive settings results in several ethical challenges that can be overwhelming or difficult for researchers and designers, especially those inexperienced in this field [5, 6, 12]. This course aims to provide concrete tools and preparation for: designing accessible and user-friendly products, services, and systems; successfully navigating power dynamics and emotionally challenging situations; and tackling ethical issues in sensitive settings. We will primarily focus on how designers and researchers can identify, act, and reflect on ethical challenges in co-designing with vulnerable user groups [6]. We will elaborate on how concepts and theories in design for dementia – such as process consent, person-centeredness, and excess disability – can be applied in other sensitive areas of HCI research. We will then let all attendees ideate and discuss how these insights support their research while receiving feedback from others and the instructors.

The first learning outcome will be the provision of **concrete insights and knowledge on how to design and develop technologies for sensitive settings**. By analyzing concrete examples from research in dementia, we will elaborate on aesthetics, interaction design, and sustainability in technology for residential care settings. The course attendees will learn and discuss how technology and products should offer non-stigmatizing aesthetics that are inviting for vulnerable users. Furthermore, we aim to provide insights into how this technology is integrated into everyday care settings and maintained beyond the course of the research project in that setting.

The second learning outcome will focus on **engaging vulnerable users and relevant stakeholders during design and research activities**. Based on existing literature combined with the instructors' experiences in dementia, we will share methods to balance asymmetric power relations in co-design settings, e.g., between caregiver and care receiver. In healthcare settings, asymmetric care relations between the patient and the professional care providers are prevalent [3]. Designers and researchers can explicitly surface these power imbalances and allow all persons involved to share their stories and have their own prerogatives reflected in the resulting design [4]. We will also focus on acting in emotionally difficult

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situations by managing expectations and goals between vulnerable users, their caregivers, family members or relatives, and researchers. And lastly, we will share experiences and tips on modifying common co-design activities to make them accessible to different stages and forms of impairment.

2 INTENDED AUDIENCE

During the course, we will discuss how the lessons learned from research in dementia apply in similar sensitive contexts, which is relevant for a broad audience working in digital design for sensitive settings. Therefore, we target designers, Master's and PhD students, or newcomers in sensitive research contexts such as aging, dementia, education, or mental health issues. We also welcome expert researchers from various fields, such as health, education, and other sensitive settings. In addition to academics, we intend to have design practitioners, workshop organizers, and facilitators present, as well as formal and informal caregivers. We have regular contact with a wide range of target groups and aim to have an inclusive conversation amongst all types of professionals.

3 PREREQUISITES

There are no prerequisites for this course other than an interest in designing for sensitive settings. This course requires no previous academic or professional experience with vulnerable user groups. We will ask the course attendees to submit their personal background and interest before the course. Next, we will divide all attendees into groups based on their professional or academic expertise to stimulate interdisciplinary interactions and knowledge exchange during the hands-on exercises.

4 CONTENT

This course will consist of one online session before the conference and one in-person session hosted at the conference venue.

4.1 Online Session: Lessons Learned from Research in Dementia

During the online session, we will provide a general overview of approaches in technology design in the context of dementia, such as in the care home or for people aging in place. We provide example cases of how to develop user-friendly, empowering, aesthetically pleasing, and non-stigmatizing products that have a meaningful impact on the everyday lives of people with dementia. We will also elaborate on the design characteristics of technology related to aesthetics, interaction, and sustainability in use. Next, we will share practical considerations (e.g., [5]), guidelines (e.g., [7]), and suitable co-creation methods (e.g., [9]) from our research. We will give attendees multiple insights into co-design practices with different groups of stakeholders in sensitive contexts and what challenges you might encounter. The attendees can expect concrete inspiration for approaches to apply for themselves, insights from literature or other domains regarding the inclusion of different target groups, and insights into the ethical challenges of this type of work.

4.2 In-Person Session: Challenges and Opportunities for Research in Sensitive Settings

During the in-person session, the course attendees will apply the insights from the previous online session to their expertise, experiences, and future research plans. We will elaborate on how these theories, methods, and products can be transferred to other sensitive settings. The attendees will apply the lessons learned in the online session to the cases in group and relate them to their personal experience in design projects or practice as each group shares their insights to feed the general discussion. For instance, concepts such as person-centered care may affect how designers construct absent users in the development of new products; ideas of process consent may be generative in understanding how to design responsible software and explainable AI; and excess disability could be positioned as a critical lens on the design of products for health and wellbeing. Within an informal and inclusive setting, the attendees can share and discuss their insights and receive feedback and tips for their future research plans.

5 PRACTICAL WORK

During the online session, there will be moments for the attendees to interact with the course instructors by asking questions or having a brief discussion. During the in-person session, the course attendees will be divided into groups of three to four people to stimulate discussion among the attendees and engagement with the course instructors, who are available to answer questions and provide feedback. We will divide the attendees into groups based on their backgrounds and expertise to stimulate interdisciplinary interactions and knowledge exchange during the exercises. The groups will discuss and work on separate cases representing different phases in a design research project. The topics will focus on different stakeholder perspectives, such as vulnerable users, their partners and family, professional caregivers, such as nursing staff or occupational therapists, and managers. Furthermore, the cases will cover different contexts, such as the home setting, daycare centers, and residential care facilities. These hands-on exercises will stimulate the attendees to reflect and discuss how to tackle ethical challenges and act in sensitive research contexts. In addition to learning guidelines and methods, this course aims to provide a platform that allows newcomers or experts in other HCI-related settings to connect, share their personal experiences, and exchange knowledge in an informal and inclusive setting.

6 INSTRUCTOR BACKGROUND

Maarten Houben is a postdoctoral researcher at the Eindhoven University of Technology in the Department of Industrial Design. His PhD research focused on the beneficial effects of everyday sounds and soundscapes in technologies for dementia. Maarten adopts a design approach for introducing and evaluating novel technologies in everyday care environments by involving key stakeholders such as people with dementia, their relatives, and care practitioners.

Minha Lee is an Assistant Professor at Eindhoven University of Technology at the Department of Industrial Design. Her research

interests include moral conflicts and emotions in relation to technology; she conducts mixed, qualitative, and quantitative research in a multidisciplinary manner. Minha Lee received her doctoral degree from the University of Eindhoven's Human-Technology Interaction and Philosophy and Ethics groups in 2021 on interactional morality.

Sarah Foley is a Lecturer in Applied Psychology and Human-Computer Interaction. Her research interests lie at the intersection of aging, care, and technology. She employs co-design and qualitative methods in her research. She completed her PhD in 2019, which examined the role of design and technology in dementia care. Current research interests include care, online communities, and qualitative methodology.

Kellie Morrissey is a Lecturer at the University of Limerick's School of Design. A psychologist by training, her main interests are in the intersection of health and politics in designing digital objects in sensitive settings. Her main research areas are participatory design approaches to engaging people with dementia in meaningful activities, with a secondary interest in design to improve women's access to appropriate and high-quality health and reproductive services.

Rens Brankaert is an Assistant Professor of Active and Healthy Ageing at Eindhoven University of Technology, Department of Industrial Design. He is also a Professor Health Innovations & Technology at Fontys University of Applied Sciences, School of Allied Health Professions. He co-leads the (TU/e) Expertise Center for Dementia & Technology. Rens has a specific interest in designing 'Warm Technology' for people living with dementia. He approaches this from a person-centered perspective and is interested in the impact of design in care practice.

7 RESOURCES

We follow methods of working with people with dementia and caregivers based on works by the instructors, such as collections of HCI research in dementia [1], concrete workshop guidelines (e.g., [7]), practical considerations for engagement with dementia [5, 11], methods on stakeholder management [2], and examples of co-creation methods (e.g., [9]). Furthermore, we will bring concrete examples of technologies and prototypes for people with dementia, such as Vita [8] or the Tumbler [9]. Originating from design research, we provide templates to think through the two learning outcomes, i.e., design and develop technologies for sensitive settings and engaging vulnerable users and relevant stakeholders during design and research activities. The templates will be visual mood boards and cards based on prior workshops that the instructors have held on designing products for people with dementia and involving vulnerable people in design research. Such workshops were the backbone of design research projects conducted with people with dementia, with concrete resources we share with the participants who will be invited to elaborate on the provided templates. During the in-person session, we stimulate interaction between the attendees, with instructors walking around to provide support and answer questions. The physical setup of the room will be similar to a workshop setting, with the tables arranged to seat groups together and a projection screen in the front of the room to share insights.

8 ACCESSIBILITY

We strive within our research to actively include people with dementia that have cognitive or physical disabilities in research. As such, we hold similar high standards of accessibility for organizing this course to be an inclusive, respectful, and diverse setting open to all. We especially welcome people who have first-hand experiences of some of the challenges or difficulties like the ones vulnerable users in sensitive settings face every day. We aim to have attendees with varying backgrounds and areas of expertise from different socio-cultural settings to tap into each other's personal experiences and allow for rich and multifaceted perspectives to emerge. We will ask every attendee to communicate their accessibility needs so we can cater to the diverse needs of all participants. We will also make use of accessibility technologies made available in the venue.

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